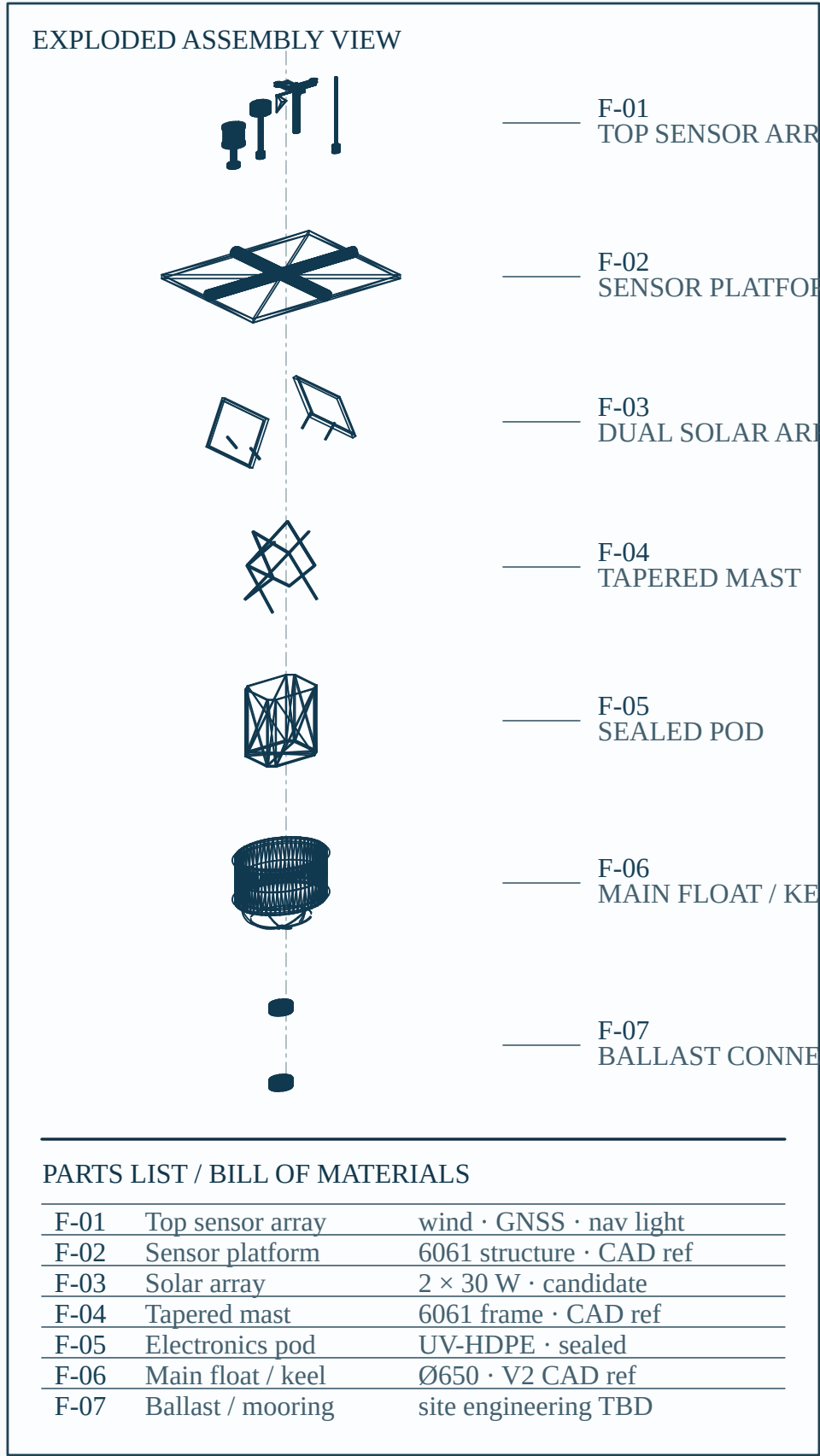
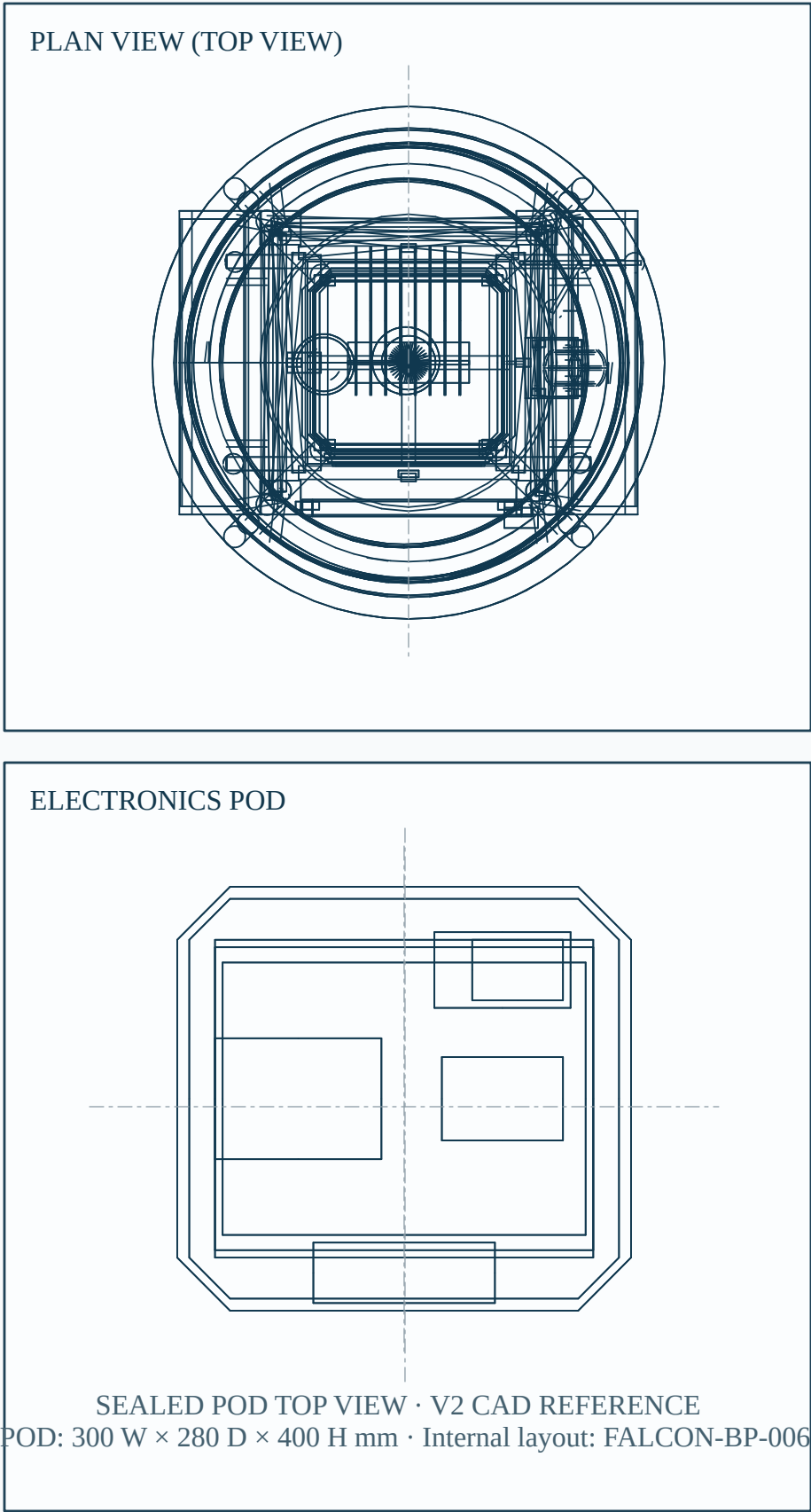
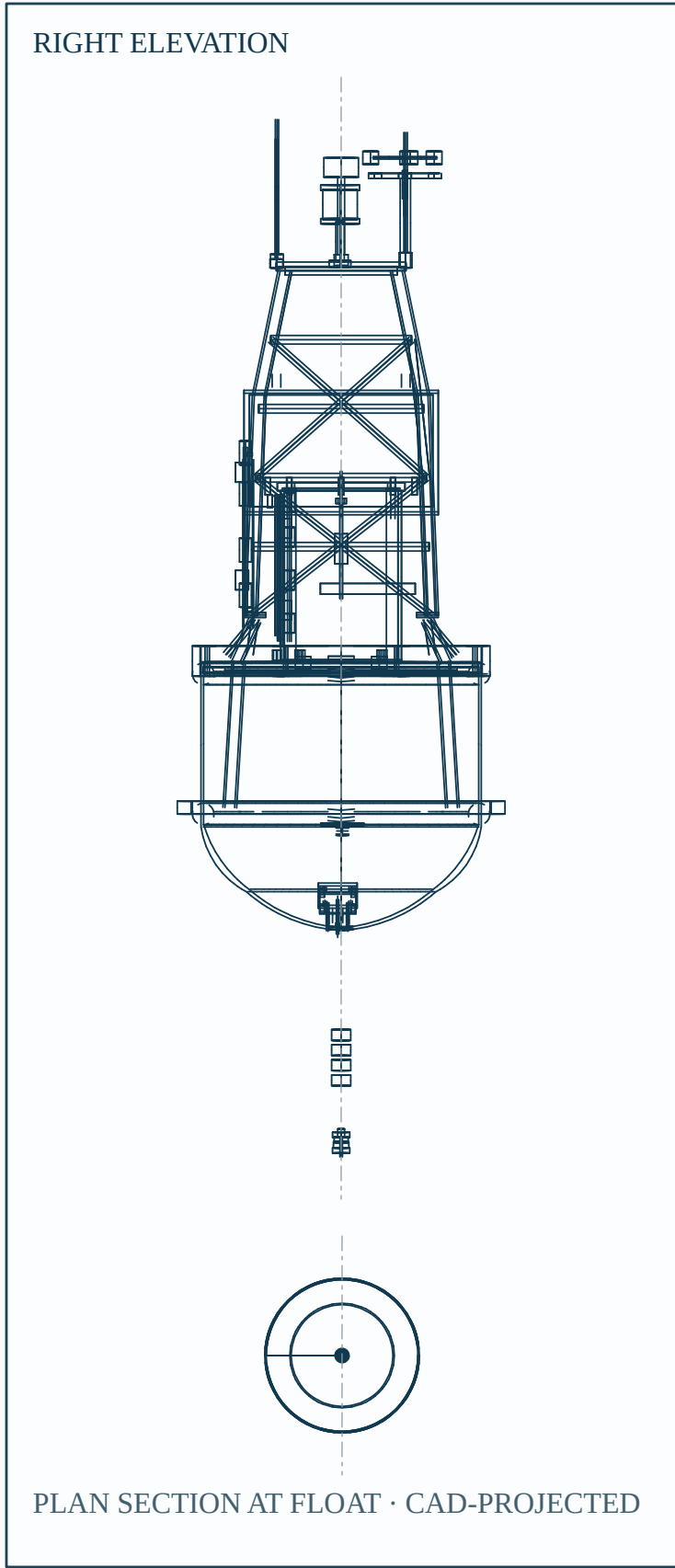
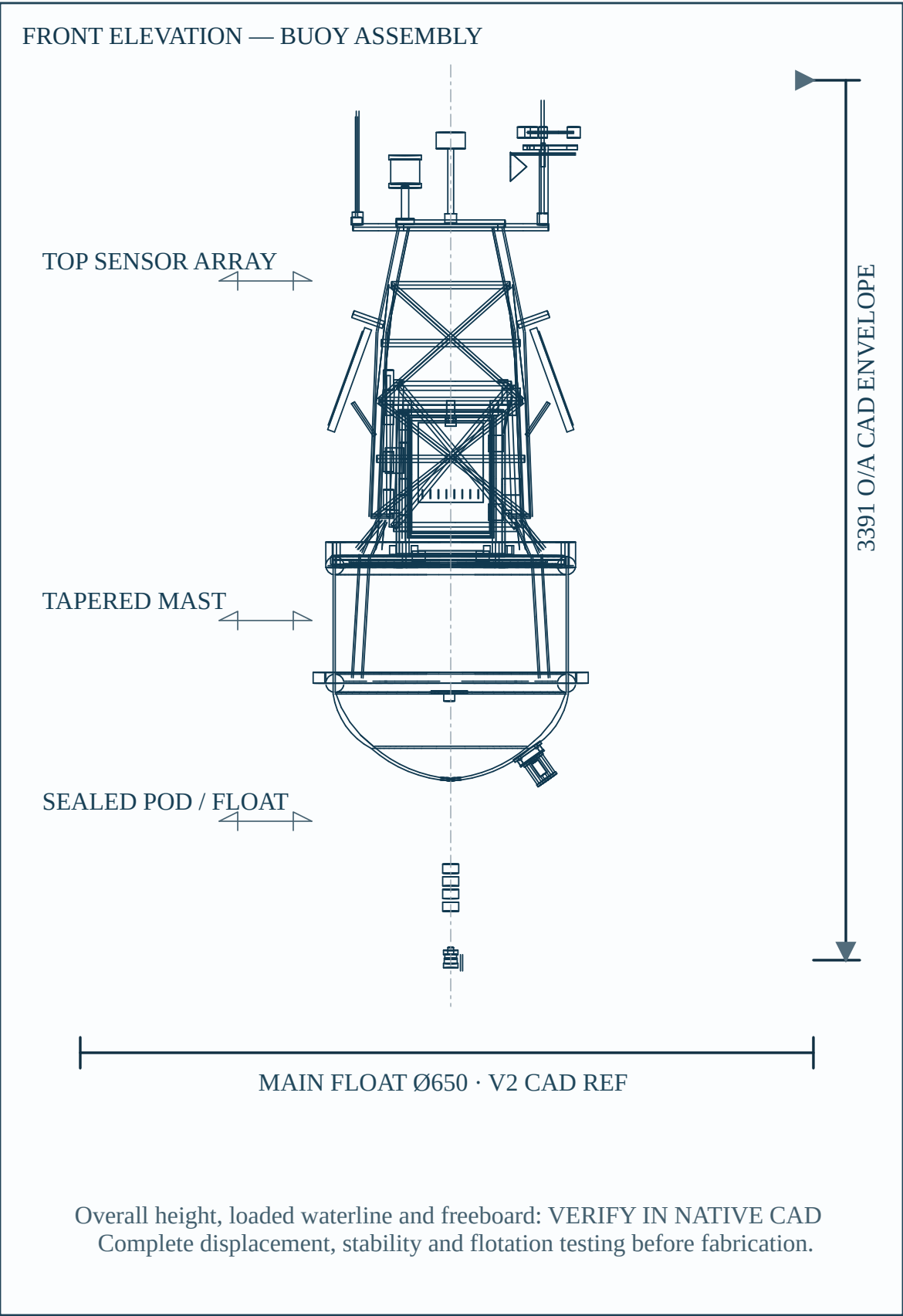
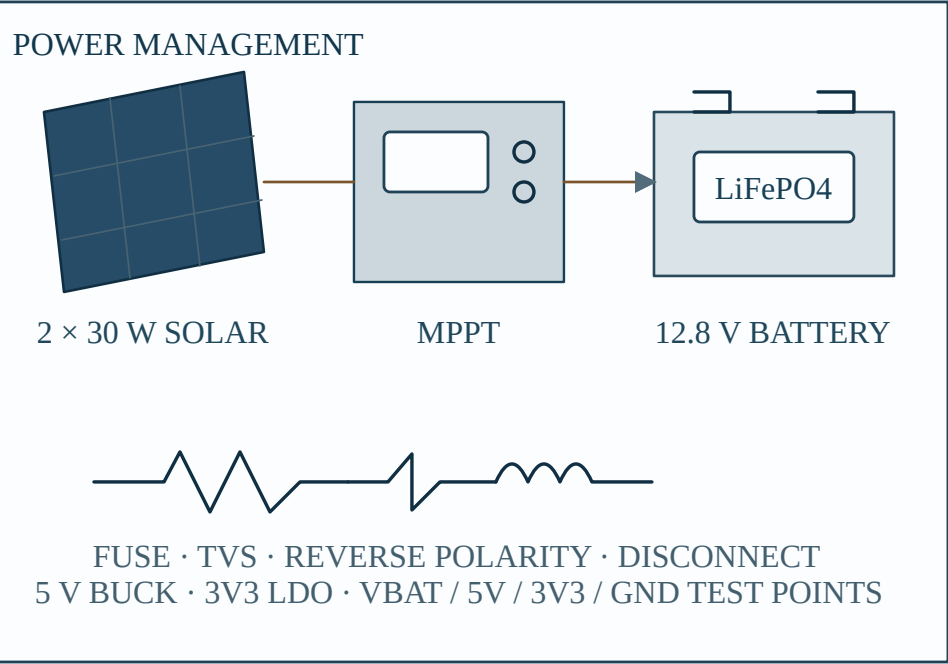




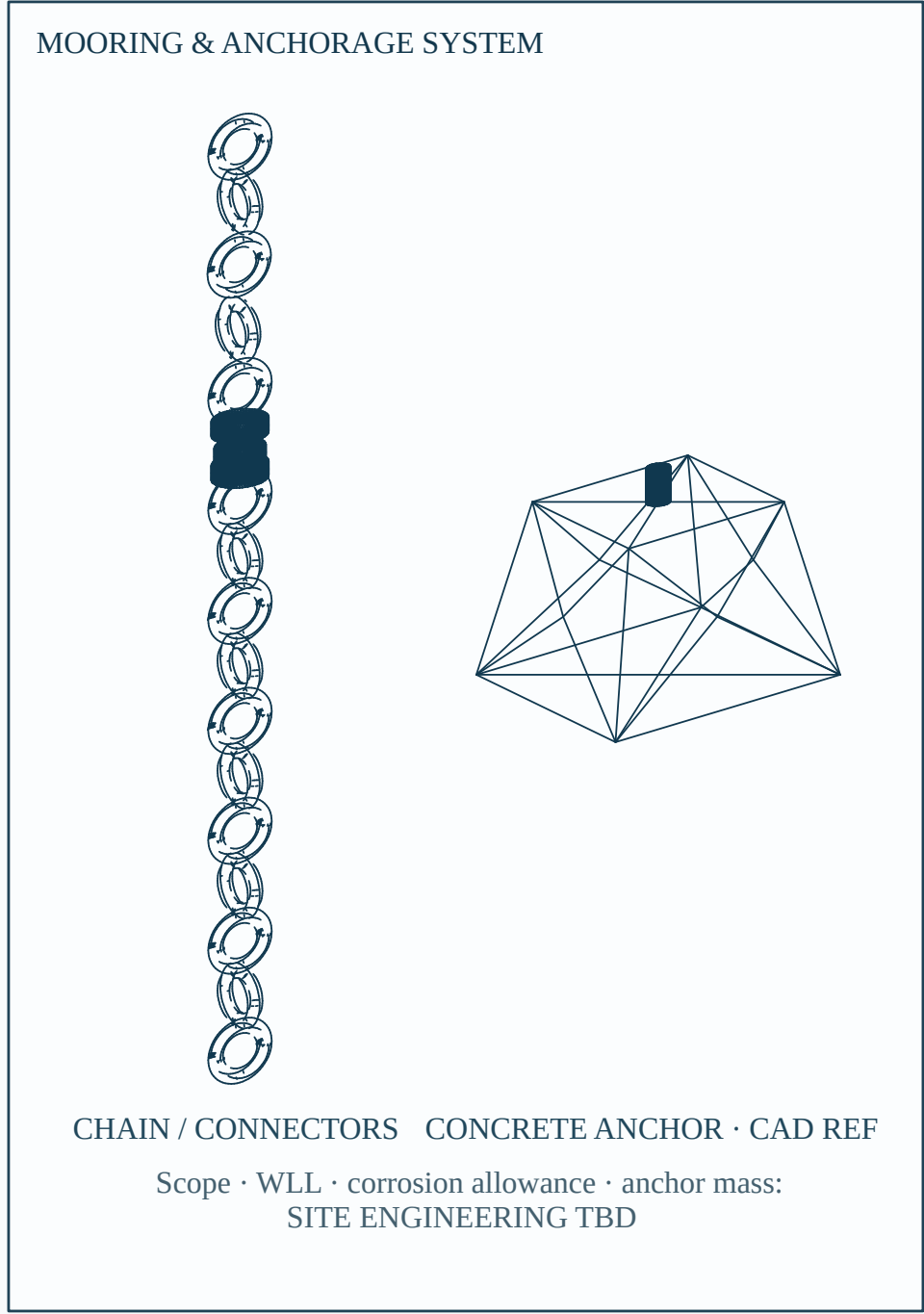
MASTER DIMENSIONS — FUSION V2 CAD REFERENCE			
ID	FEATURE	FUSION DIMENSION	ENGINEERING STATUS
D01	Complete buoy envelope	760 W × 760 D × 3391 H mm	CAD-export envelope; excludes site mooring run
D02	Main float / keel	Ø650 × 620 H mm	380 upper body + 240 tapered keel; 6 mm wall
D03	Tapered mast frame	805 H mm envelope	800 mm design height; 4 × Ø32 legs
D04	Sealed electronics pod	300 W × 280 D × 400 H mm	8 mm UV-HDPE wall; 18 mm service lid
D05	Dual solar modules	2 × 450 × 300 × 20 mm	30 W each; 20° outward tilt
D06	Pressure sensor / guard	Ø24 × 42 / Ø64 × 72 mm	Mounting depth and bracket remain approval items
D07	Ballast rail / plates	Ø40 × 500; 4 × Ø220 × 25 mm	Final mass, CG and righting test required
D08	Concrete anchor reference	650/450 square × 500 H mm	Nominal 367 kg; site engineering approval require
DO NOT SCALE DRAWING — VERIFY FINAL FABRICATION DIMENSIONS IN THE NATIVE FUSION PARAMETRIC MODEL			

COMPONENT / INTERFACE NOTES	
S01 Bar02 R2	— pressure time series for estimated wave height; service limitation must be resolved.
S02 Ultimate GPS PID 746	— position/geofence; UART GPIO16/17; field scatter test required.
S03 SEN-15901	— wind pulse GPIO25 and vane through ADS1115 A0; marine durability unqualified.
S04 Celsius R2	— deployment water-temperature candidate; exact harness/address must be verified.
H01/H02 MCP9808 + INA260 ×2	— enclosure temperature and battery/solar electrical health.
C01/C02 ESP32-DevKitC V4 + SIM7600G-H candidate	— LTE interface and peak current not frozen.
EXCLUDED BNO085, load-cell/HX711, onboard Orange Pi and salinity sensor.	

GENERAL NOTES	
1.	CAD linework is projected from PROJECT-FALCON-V2.glb; do not scale this sheet.
2.	Project FALCON estimates wave height from submerged pressure time series at the shore Bay Station.
3.	Orange Pi / Bay Station computer remains ashore; no single-board computer is installed on the buoy.
4.	Verify purchased components, pin 1, voltages, connector orientation, sealing, loads and calibration.
5.	Complete flotation, stability, structure, thermal, ingress, corrosion, power and mooring reviews before build.
REFERENCE / NOT FOR FABRICATION	



WIRING SCHEMATIC OVERVIEW		
	ESP32	PROTECTED I/O
I2C SDA / SCL		
GPS UART RX / TX		
WIND PULSE / ADC		
LTE UART / USB		
3V3 / 5V / GND		
FUNCTIONAL OVERVIEW ONLY Use verified KiCad schematic and harness for fabrication		



DRAWING CONTROL		DRAWING TITLE ASSEMBLY & SUBSYSTEMS	
PROJECT	FALCON-01	REVISION	P5
DRAWING NO.	FALCON-BP-000	DATE	2026-09-04
SCALE	NTS	STATUS	REFERENCE / NOT FOR FABRICATION
SOURCE	V2 GLB + CONTROLLED DOCS		